

Case Study: Distinguishing AI and Real Images



The Scenario:

You are the head of the FBI. The President of the United States has called you into the Oval Office to brief you on a national emergency. Images can no longer be used as evidence in court, as defense attorneys argue any images that implicate their clients in criminal activity are AI-generated, while prosecutors use AI-generated images to press charges against innocent people. The President has assigned you a critical task: Can we determine, with a high level of accuracy, whether an image is real or AI-generated? Our nation's criminal justice system depends on it.

The Mission:

Your mission is to build an image classifier that can accurately predict whether or not an image is real or AI-generated beyond random guessing. You will use labeled real and AI images to train this model for use on a test set of unseen images. You don't need to build a 100% accurate classifier, as even 80% test set accuracy can help lead to more reliable convictions and acquittals.

You will be responsible for the following:

- Convert the provided images into pixel value tensors to be used as inputs in a ConvNeXT classifier.
- Decide how many epochs for which to train the model. More epochs may take longer to train and test, but may improve accuracy.
- Decide whether to simply predict whether an image is real or AI-generated, or to predict the exact model that generated the image. Doing the latter may provide additional evidence that an image is AI-generated if you can identify which model produced it, though it may be less accurate overall.
- If you choose a binary classification model, balance precision and recall. Is it more damaging to have AI-generated images classified as real, or real images classified as AI-generated?
- Use a confusion matrix to visualize your test set predictions